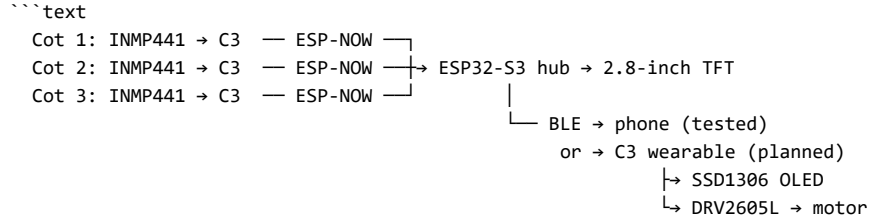
CryNet — USB Test and Complete-System Wiring Guide

Revision: 2 November 2026. Read alongside `HARDWARE_COMPONENTS.md`.

> **Important:** The node + hub signal wiring in section 2 reflects the working prototype. The wearable, battery power, node LEDs and battery sensing are **not implemented or verified**. Their exact part-to-part power wiring and GPIO choices cannot safely be finalised until the actual board and charger modules, cell and motor are selected. Section 3 is a proposed connection architecture, not permission to wire an unspecified lithium charger. This academic prototype is not a certified infant-monitoring device and must not replace supervision. Keep cables, cells, modules and small parts out of reach of the infant.

1. System picture



No audio or signal wires connect the cot nodes to the hub: the link is wireless. Each assembly has its own local power and ground. The node, hub and wearable do not need grounds connected to each other.

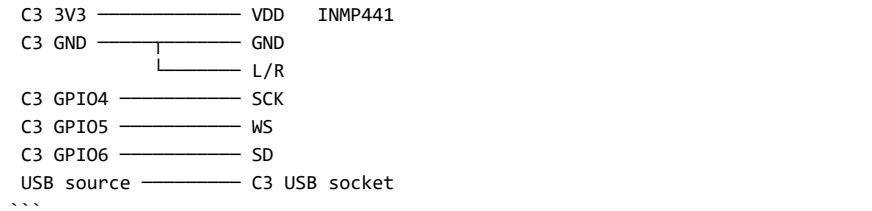
2. Wiring A — USB-powered bench test

Before wiring: disconnect power; identify actual board/module pin labels; use a multimeter to check for shorts and verify the display module voltage. These are board pin labels, not physical header positions. Do not assume every breakout uses the same pin order.

2.1 Cot node: ESP32-C3 + INMP441

Repeat the same connections for each node. The current real-board test used one node; three-node simultaneous operation is not yet verified.

From	To	Note
C3 **3V3**	INMP441 **VDD / 3V3**	3.3 V only; not USB 5 V.
C3 **GND**	INMP441 **GND**	Common local ground.
C3 **GND**	INMP441 **L/R**	Select left I²S channel.
C3 **GPIO4**	INMP441 **SCK**	I²S bit clock.
C3 **GPIO5**	INMP441 **WS**	I²S word select.
C3 **GPIO6**	INMP441 **SD**	Microphone data to C3.
Regulated USB supply or computer	C3 **USB connector**	Use a compatible data cable for flashing/serial.
Optional 100 nF capacitor	Between mic **VDD and GND**	As close as practical to microphone; check existing breakout decoupling. Ceramic is non-polar.



The old PDF's GPIO5/6/7 microphone assignment does **not** match the current firmware. C3 GPIO4 is already used by SCK and must not also be connected to the old red status LED.

2.2 Hub: ESP32-S3 + 2.8-inch ILI9341

The project log records the following signal and 3.3 V connections on the existing display. Match the actual module's labels and confirm its supply and backlight circuit before copying it to a different TFT.

From S3	To display	Note
3V3	**VCC**	Recorded working connection for the existing module; do not assume another module has identical power requirements.

GND	**GND**	Local common ground.
GPIO10	**CS**	Chip select.
GPIO11	**MOSI / SDI**	SPI data to TFT.
GPIO12	**SCK / CLK**	SPI clock.
GPIO13	**DC / D-C**	Data/command.
GPIO14	**RST / RESET**	Display reset.
GPIO4	**BL / LED control**	Recorded on the existing board. Check the module's backlight drive current and circuit; do not drive an unknown bare LED backlight from a GPIO.
Not connected	**MISO / SDO**	The current driver does not read TFT data.
Regulated USB supply or computer	S3 **USB connector**	Match the actual connector used for programming.

No external 100 Ω backlight resistor should be added blindly; use the specific module's circuit. Touchscreen and SD-card pins, if present, are not used by the current display firmware. The S3's BOOT button is used by the existing UI; it is already on the development board.

2.3 USB wearable bench wiring — proposed, not yet built

The firmware and exact GPIO assignments do not yet exist. The following is a **connection list**, not a tested pin map. Select free C3 I²C pins after checking the actual board, USB and boot-strapping constraints; use the same two bus lines for both modules.

From	To	Condition
C3 **GND**	OLED **GND** and DRV2605L **GND**	Common local ground.
C3 **3V3**	OLED **VCC**	Only if the specific OLED breakout accepts 3.3 V.
Appropriate regulated supply	DRV2605L breakout **VIN/VDD**	Follow the actual breakout's voltage requirements; do not assume its VIN and logic pins are identical to the IC pins.
Selected C3 I²C SDA	OLED **SDA** and driver **SDA**	Bus pull-ups must go to a safe 3.3 V logic rail. Check existing pull-ups before adding any.
Selected C3 I²C SCL	OLED **SCL** and driver **SCL**	Same. Verify address compatibility.
Driver **OUT+ / OUT−**	Compatible motor's two terminals	Use the specific breakout's labels and motor requirements. Do not connect either motor lead to 3V3 or ground as in the old MOSFET circuit.
USB source	C3 USB socket	Supply capability and driver power arrangement must be checked.

Use the selected breakout's documentation for EN, I/O supply, power, pull-ups and any other control pins; these vary by board. DRV2605L uses an H-bridge, unlike the old low-side MOSFET design. No separate MOSFET or old motor flyback-diode wiring is used for this driver. See [TI DRV2605L datasheet](https://www.ti.com/lit/ds/symlink/drv2605l.pdf). Test firmware is needed before this wearable can receive BLE alerts.

3. Wiring B — proposed battery-powered complete system

3.1 Connections that remain the same

For each of the three nodes, retain **all six microphone connections** in section 2.1. For the hub, retain the TFT signal and ground connections in section 2.2. For the wearable, use only a finalised and tested version of section 2.3. Only the **power source** and proposed indicators/battery sensing change; never join all five batteries together.

3.2 Power architecture for each separate unit

```text
 USB charging input
        │
        ▼
 [charger + properly designed power-path/load sharing] ↔ [protected 1-cell battery]
        │ regulated/selected system output, as the module specifies
        ▼
 [suitable regulated converter, if needed] → [power switch as designed]
        │                                        │
        ├── appropriate board power input        └── local loads, if designed
        └── local ground / return
```

This diagram deliberately does not name charger terminals: TP4056 boards vary, and `BAT`, `OUT`, and protection pins are not universally interchangeable. A common charger/protection module is not necessarily a load-sharing circuit. Choosing terminal-by-terminal wiring without a specific module, current rating and charging-use requirement could cause incorrect charging, backfeeding, resets or fire.

For each of the three nodes and the hub, the BOM proposes a protected 1-cell battery and a 5 V boost if the selected board uses its 5 V input. The wearable's suitable voltage/converter remains undecided. A lithium cell can be **4.2 V fully charged**, so do not connect it to a 3.3 V rail or GPIO; do not apply 5 V to 3V3. The converter's output must remain within the board input specifications under measured peak loads and low cell voltage. A 100 µF bulk capacitor is a **candidate**, not an unconditional connection: confirm location, voltage rating and stability requirements in the

actual converter datasheet, then connect + to the intended rail and – to local GND only if appropriate. Do not connect bench USB and a battery supply simultaneously without a designed, verified power path preventing backfeed.

3.3 Node battery voltage measurement — proposed

Only after confirming the selected C3 ADC pin, its voltage/attenuation limits and firmware:

```text
 battery positive (cell side) ── 100 kΩ ─┬─┬── selected ADC pin
                                         ├── 100 kΩ ── GND
                                         └── 100 nF ── GND
```

At 4.2 V cell voltage the nominal ADC voltage is 2.1 V. Account for component tolerances, absolute/operating limits and calibration. The divider continuously draws current; consider a switched design. **No GPIO is assigned here:** GPIO4–6 are occupied by the microphone, and strapping/USB pins require care. Battery measurement and low-battery reporting are not implemented in the current node firmware.

3.4 Node status LEDs — proposed

For each LED, a common active-high arrangement is `selected GPIO → 330 Ω → LED anode; LED cathode → GND`, but **do not solder to a specific pin yet**. Select two free GPIOs after checking the particular board's boot-strapping, USB and other functions and implement the corresponding firmware. The old PDF's GPIO4 red LED conflicts with SCK; its GPIO9 blue LED uses a strapping pin and must not be copied without a boot-state analysis. Verify LED current and resistor value for the selected parts.

3.5 Wearable battery and haptics

The planned DRV2605L motor output and I²C connections are as in section 2.3, subject to selection of compatible actual boards and motor. Its battery, charger, protection, load-sharing, conversion and switch must be designed as one circuit. The proposed 500 mAh is only a capacity idea: measure current and choose the charge rate per the actual cell. Never assume a 1 A TP4056 default is suitable. The IRLML2502 and its gate resistors/diode belong to the alternative MOSFET drive and are **not wired into the DRV2605L path**.

4. Checks before power-up

1. With power disconnected, check continuity and make sure no 3.3 V or 5 V rail is shorted to ground. Inspect vero-board copper tracks, intentional cuts and solder bridges.
2. Confirm the exact module labels, voltage ratings, polarity, cell protection and regulator settings. Check converter output with a meter **before** connecting its load.
3. For initial testing use a current-limited bench supply where possible. Test one assembly at a time; avoid unattended charging.
4. Test the known USB node-to-hub setup first. The wearable and additional nodes require their own implementation and verification.
5. Before using battery and USB together, verify the actual charger/load-sharing and backfeed behaviour. Secure cells against puncture/shorts and use appropriate enclosures.

5. Information needed to finish exact battery wiring

Obtain the manufacturer/model and clear front/back labels or schematics of the five selected charger/power-path modules, converters, cells and board variants; the exact OLED, DRV2605L breakout and motor; and whether operation while charging is required. Measure peak hub and node currents. Then select the battery and charge currents, free GPIOs and pull-ups, produce a module-specific point-to-point schematic and validate before assembly. Until then, this guide cannot honestly claim exact full-system battery wiring.

Evidence: `firmware/node/node_crynet/node_crynet.ino` (GPIO4/5/6); `firmware/hub_infer/components/hub_display/hub_display.c` (TFT pins); `SYSTEM_UPDATE.md` (working display wiring and hardware tests); `HARDWARE_COMPONENTS.md` (proposed BOM).